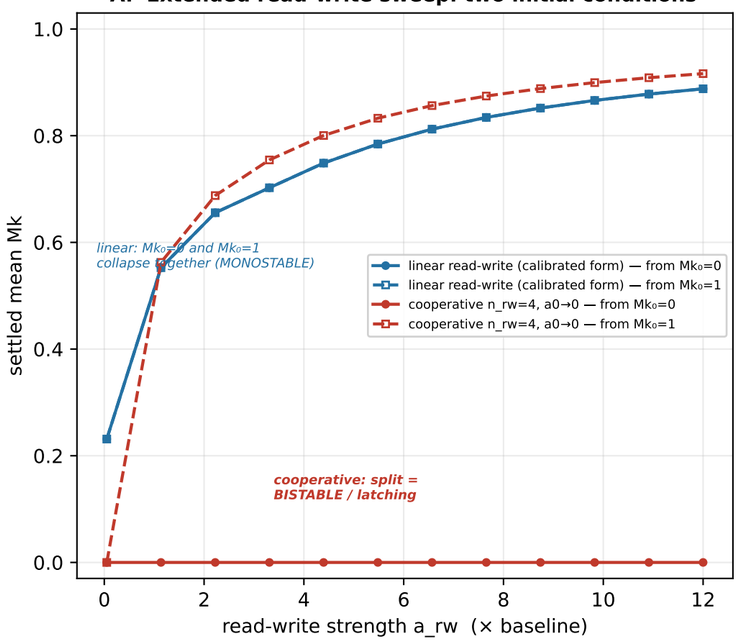
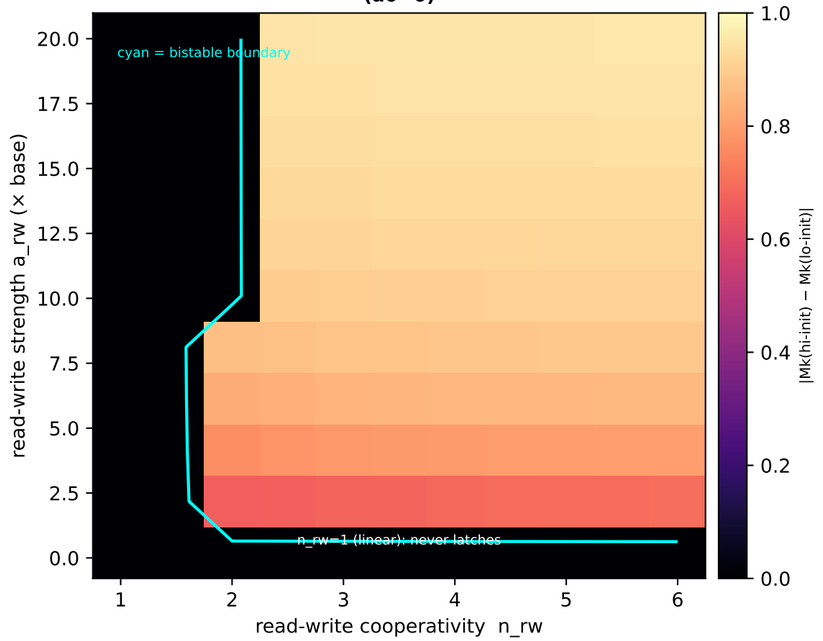
Possibility space of the *Ccnd1* H3K27me3 mark — extended bounds + structural knobs (v44.1 dynamics; not data-constrained)

Calibrated architecture is monostable (buffered). Cooperative read-write (n_rw≥2) + suppressed nucleation (a0→0) unlocks a BISTABLE latching switch coupled to cell-cycle arrest — a reachable behavior the data-fit model does not use.

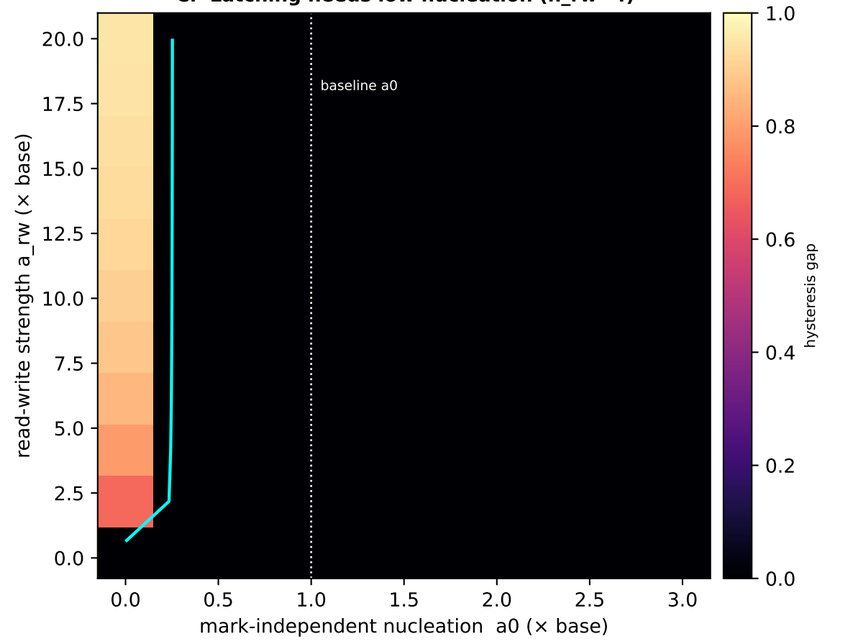
A. Extended read-write sweep: two initial conditions



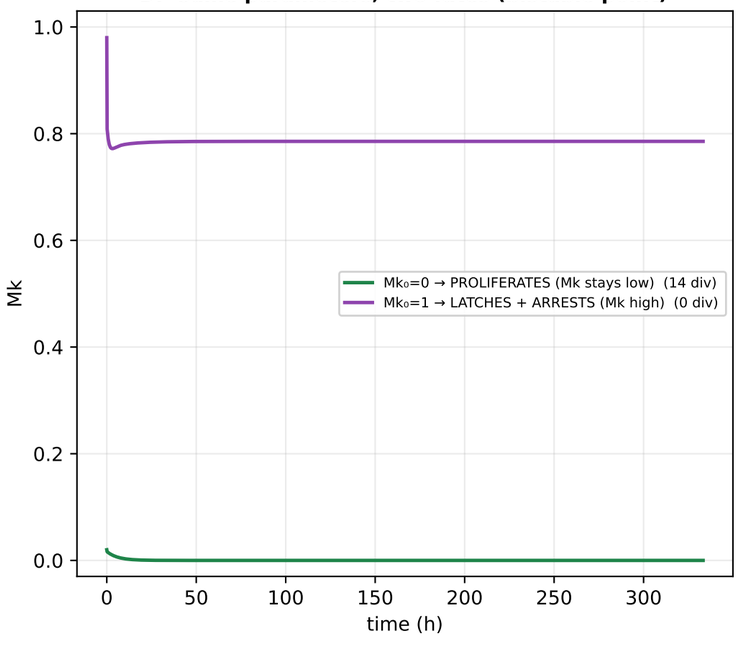
B. Latching (hysteresis gap) vs cooperativity × strength (a0=0)



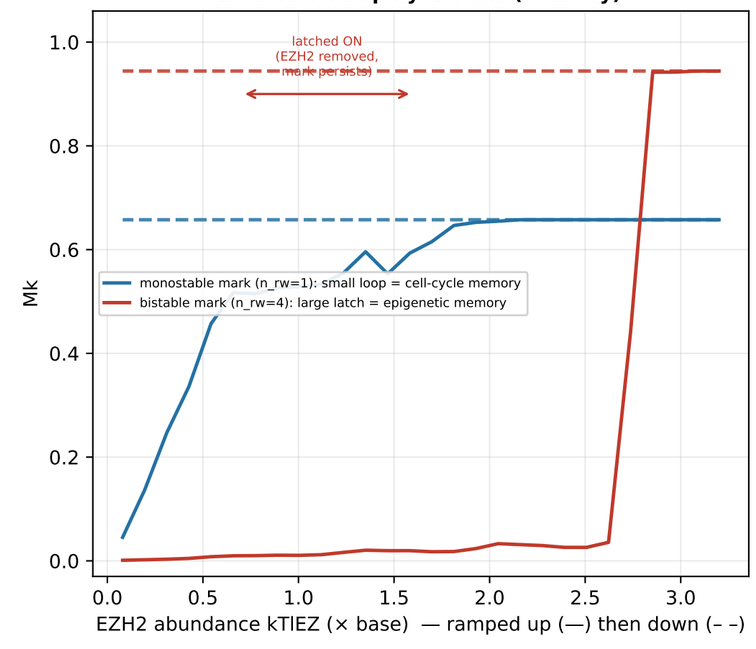
C. Latching needs low nucleation (n_rw=4)



D. Same parameters, two fates (bistable point)



E. Driver-ramp hysteresis (memory)



F. Regime map (a0=0)

